

Grade 6 Accelerated
Unit 10
Lesson 7 Homework

Name Answer Key
Date _____
Period _____

A group of students challenged each other to see how long it took them to finish the same math problem. The students want to see who can solve the problem correctly the fastest. The times of the students are listed in seconds is represented in the stem and leaf plot.

Stem	Leaf
0	80
1	02 15 45 59 60
2	04 32 45
3	33

Key: 1|02 = 102

- Find the mean and median of the data set.
Median: 159.5 seconds
Mean: $\frac{1775}{10} = 177.5$ seconds
- The range of the data set is **$333 - 80 = 253$** .
- The IQR of the data set is **$232 - 115 = 117$** .

A different group of students in the same class timed themselves to see how long it took them to solve the same math problem and recorded their time in seconds.

{134, 122, 111, 90, 101, 103, 143, 60, 45, 100}

- Find the mean and median of the data set.
Median: 102 seconds
Mean: $\frac{1009}{10} = 100.9$ seconds
- The range of the data set is **$143 - 45 = 98$** .
- The IQR of the data set is **$112 - 90 = 22$** .

For questions 7–10, determine whether each statement is true or false. Explain your reasoning.

7. The first group of students had a lower median than the second group of students.

False. The median of the first group was 159.5 seconds which is higher than the second group's, 102 seconds.

8. The center of group 1 is similar to the center of group 2.

False. The center measures for group 1 (177.5, 159.5) were both larger than the center measures for group 2 (100.9, 102).

9. The second group of students has a larger range than the first group of students.

False. The range of the second group, 98, is much smaller than the range of the first group, 253.

10. The first group of students is more consistent with their times than the second group.

False. Explanations may vary.

Group 2's mean and median were close in value, and group 2's range and IQR were smaller. Therefore, group 2 was more consistent with their times.